# Ho-min Park

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## ABOUT ME

Computer Science Engineering PhD with deep expertise in machine learning, spanning from foundational concepts to state-of-the-art architectures including CNNs, Transformers, GANs, and Diffusion models. Designed and delivered a comprehensive 20-lecture ML curriculum over 4 years through the AI Vacation School (AIVS) program, creating original educational materials with lecture slides, hands-on Colab notebooks, and step-by-step Excel worksheets for algorithm understanding. Experienced in evaluating and iteratively improving ML educational content based on learner feedback and evolving research trends. Authored 7 peer-reviewed publications applying ML across diverse domains (medical imaging, bioinformatics, affective computing, environmental science), providing broad perspective for assessing ML lecture materials across application areas.

## ML EDUCATION & CURRICULUM DESIGN

**AI Vacation School (AIVS) – Founder & Lead Instructor** 2021 - 2025
**Ghent University Global Campus**

- ✓ Founded and led a volunteer-driven ML bootcamp (100+ hours per session) for undergraduate students over 7 sessions across 4 years, iteratively refining content based on learner feedback and emerging research trends. [Introduction, Lecture Slides]
- ✓ Designed a 20-lecture curriculum spanning the full ML pipeline: regression, MLPs, CNNs, RNNs, Transformers, LLMs, GANs, Diffusion models, unsupervised learning, and XAI (SHAP).
- ✓ Created three types of original materials per topic: presentation slides, Google Colab notebooks, and Excel worksheets for step-by-step manual computation of ML algorithms.
- ✓ Developed a three-stage pedagogical framework (pre-session self-study, on-site intensive training, independent research project) with level-based progression (Beginner, Advanced, Advanced Plus).
- ✓ Mentored over 10 entry-level AI researchers with follow-up seminars, portfolio development guidance, and recommendation letters.

## EDUCATION

**Ph.D. in Computer Science Engineering** 2018 - 2025
**Ghent University, Belgium**

- ✓ Dissertation: “Advancing Detection through Deep Learning for Industrial, Environmental, and Health Applications”
- ✓ Advisors: Prof. Wesley De Neve, Prof. Arnout Van Messem

**M.S. in Computer Engineering** 2016 - 2018
**Ajou University, Suwon, South Korea**

- ✓ Thesis: “Cited count prediction modeling from large citation network"

**B.S. in Computer Science and Engineering** 2010 - 2016
**Ajou University, Suwon, South Korea**

- ✓ Major in Computer Science and Engineering

## SKILLS

### ML & AI Expertise

- ✓ *Architectures:* CNN, RNN, LSTM, GRU, Transformer, GAN, Diffusion Models, Multi-layer Perceptrons
- ✓ *Techniques:* Supervised/Unsupervised Learning, Transfer Learning, Feature Engineering, Model Explainability (SHAP, CAM)
- ✓ *Domains:* Medical Imaging (MRI), Bioinformatics (CRISPR-Cas), Affective Computing, Environmental Science, Industrial Systems

### Educational Tools & Platforms

- ✓ *Curriculum Design:* 4 years of ML curriculum development, iterative improvement based on learning outcomes
- ✓ *Teaching Platforms:* Google Colab, Jupyter Notebook, GitHub Pages, reveal.js-based slide systems
- ✓ *MLOps & Development:* PyTorch, PyTorch Lightning, Weights and Biases (WandB), Python

### Languages

**Korean**

- ✓ Native speaker.

**English**

- ✓ *Writing:* Demonstrated through technical documentation and research publications.
- ✓ *Reading:* Evident through the comprehension of technical and academic literature.
- ✓ *Speaking and Listening:* Applied in collaborative environments with international teams.

## PUBLICATIONS

### Journal Articles

- ✓ Park, H., Kim, G., Oh, J., Van Messem, A., & De Neve, W. (2024) Interpreting stress detection models using SHAP and attention for MuSe-Stress 2022. *IEEE Transactions on Affective Computing*, **15**(01): 1-17. DOI: 10.1109/TAFFC.2024.3488112.
- ✓ Park, H., Won, J., Park, Y., Anzaku, E. T., Vankerschaver, J., Van Messem, A., De Neve, W., & Shim, H. (2023) CRISPR-Cas-Docker: Web-based in silico docking and machine learning-based classification of crRNAs with Cas proteins. *BMC Bioinformatics*, **24**(1): 167. Springer.
- ✓ Park, H., Park, Y., Berani, U., Bang, E., Vankerschaver, J., Van Messem, A., De Neve, W., & Shim, H. (2022) In silico optimization of RNA-protein interactions for CRISPR-Cas13-based antimicrobials. *Biology Direct*, **17**(1). DOI: 10.1186/s13062-022-00339-5.
- ✓ Park, H., Park, Y., Vankerschaver, J., Van Messem, A., De Neve, W., & Shim, H. (2022) Rethinking protein drug design with highly accurate structure prediction of anti-CRISPR proteins. *Pharmaceuticals*, **15**(3). DOI: 10.3390/ph15030310.
- ✓ Park, H., Park, S., de Guzman, M. K., Baek, J. Y., Cirkovic Velickovic, T., Van Messem, A., & De Neve, W. (2022) MP-Net: Deep learning-based segmentation for fluorescence microscopy images of microplastics isolated from clams. *PLoS ONE*, **17**(6): e0269449. DOI: 10.1371/journal.pone.0269449.

### Conference Articles

- ✓ Park, H., Yun, I., Kim, M., Nguyen, K. T., Van Messem, A., & De Neve, W. (2024) Interpretable Rotator Cuff Tear Diagnosis Using MRI Slides with CAMscore and SHAP. *Medical Imaging 2024: Computer-Aided Diagnosis*, Vol. 12927: 683-693. SPIE.

- ✓ Singh, A. K., Kim, G., Kim, J., Park, H., Choi, B. J., & De Neve, W. (2024) Exploring Multimodal Approaches and Fusion Methods for CEO Social Attribute Prediction in 2024 MuSe-Perception. *Proceedings of the 5th on Multimodal Sentiment Analysis Challenge and Workshop: Social Perception and Humor*, 36–44. Melbourne VIC, Australia: ACM. DOI: 10.1145/3689062.3689081.
- ✓ Park, H., Kim, G., Van Messem, A., & De Neve, W. (2023) MuSe-Personalization 2023: Feature engineering, hyperparameter optimization, and transformer-encoder re-discovery. *Proceedings of the 4th Multimodal Sentiment Analysis Challenge and Workshop: Mimicked Emotions, Humour and Personalisation*, 89–97. Ottawa ON, Canada: ACM. DOI: 10.1145/3606039.3613104.
- ✓ Park, H., Yun, I., Kumar, A., Singh, A. K., Choi, B. J., Singh, D., & De Neve, W. (2022) Towards multimodal prediction of time-continuous emotion using pose feature engineering and a transformer encoder. *Proceedings of the 3rd International on Multimodal Sentiment Analysis Workshop and Challenge*, 47–54. DOI: 10.1145/3551876.3554807.
- ✓ Baek, J. Y., de Guzman, M. K., Park, H., Park, S., Shin, B., Cirkovic Velickovic, T., Van Messem, A., & De Neve, W. (2021) Developing a segmentation model for microscopic images of microplastics isolated from clams. *Pattern Recognition, ICPR International Workshops and Challenges, Proceedings, Part VI*, 12666: 86–97. Cham: Springer. DOI: 10.1007/978-3-030-68780-9_9.
- ✓ Park, H., Kang, B., Van Messem, A., & De Neve, W. (2021) 3-D deep learning-based item classification for belt conveyors targeting packaging and logistics. *Pattern Recognition, ICPR International Workshops and Challenges, Proceedings, Part VI*, 12666: 578–591. Cham: Springer. DOI: 10.1007/978-3-030-68799-1_42.
- ✓ Kang, H., Park, H., Ahn, Y., Van Messem, A., & De Neve, W. (2021) Towards a quantitative analysis of class activation mapping for deep learning-based computer-aided diagnosis. *Medical Imaging 2021: Image Perception, Observer Performance, and Technology Assessment, Vol. 11599*, SPIE. DOI: 10.1117/12.2580819.
- ✓ Park, H., Van Messem, A., & De Neve, W. (2020) Item measurement for logistics-oriented belt conveyor systems using a scenario-driven approach and automata-based control design. *IEEE 7th International Conference on Industrial Engineering and Applications (ICIEA 2020)*, Bangkok and Phuket.
- ✓ Park, H., Van Messem, A., & De Neve, W. (2019) Box-Scan: An efficient and effective algorithm for box dimension measurement in conveyor systems using a single RGB-D camera. *The 7th IIAE International Conference on Industrial Application Engineering*, Kitakyushu, Japan.

## FUNDED PROJECTS

**Depression Diagnosis & Drug Adherence Research (2022 - 2025)**

- ✓ Developed advanced models for the MuSe-Stress and MuSe-Personalization challenges.
- ✓ Implemented Transformer Encoders and novel feature engineering techniques.
- ✓ Achieved 2nd place in MuSe-Personalisation 2023 and 3rd place in MuSe-Stress 2022.
- ✓ Contributed to the field of affective computing with interpretable AI models.
- ✓ **Funding:** National Research Foundation of Korea (Grant No.: NRF-2020K1A3A1A68093469), Department of Biotechnology in India (Grant No.: DBT/IC-12031(22)-ICD-DBT), and Ghent University Global Campus

**Environmental Monitoring Project (2020 - 2023)**

- ✓ Led development of MP-Net, an AI system for automated microplastics detection
- ✓ Designed and implemented a microplastics annotation tool for dataset creation
- ✓ Managed collaboration between environmental scientists and AI engineers

- ✓ **Funding:** Special Research Fund of Ghent University (Grant No.: 01N01718) and Ghent University Global Campus

**Smart Packaging System Development (2018 - 2021)**

- ✓ Developed and patented two AI-powered measurement systems for industrial conveyor belts
- ✓ Led implementation of 3D deep learning-based item classification system
- ✓ Designed integrated size and type monitoring system
- ✓ Collaborated with industry partners to deploy solutions in real factory environments
- ✓ **Funding:** Ministry of Trade, Industry, and Energy (MOTIE), Korea (Project No.: 10077285)

## PATENTS

- ✓ Park, H., & De Neve, W. (2020) An apparatus and method for measuring the size of high-speed objects and classifying their type on conveyors using light curtain sensors. Korean Intellectual Property Office. Application no.: 10-2020-0045098, Registration no.: 10-2273758.
- ✓ Park, H., & De Neve, W. (2018) Apparatus and method for measuring the size of high-speed boxes on conveyors using an RGB-D camera. Korean Intellectual Property Office. Application no.: 10-2018-0148118, Registration no.: 10-2066862.

## AWARDS & CERTIFICATIONS

**Awards**

- ✓ **Second Place**, MuSe-Personalisation Challenge 2023, awarded for multimodal sentiment analysis contributions.
- ✓ **Third Place**, MuSe-Stress Challenge 2022, awarded for contributions to multimodal sentiment analysis.
- ✓ **Best Presentation Award**, International Conference on Industrial Application Engineering (IIAE), 2019, for effective research communication and presentation.

**Certifications**

- ✓ **TOEIC** – Score: 860 points, Date: September 7, 2024
- ✓ **OPIc** – Level: Intermediate High (IH), Date: September 17, 2024
- ✓ **Information Processing Engineer** – Certification No.: 15202100118F, Date: July 31, 2015
- ✓ **Information Processing Industrial Engineer** – Certification No.: 12642002187H, Date: November 30, 2012
- ✓ **Information Equipment Operation Craftsman** – Certification No.: 06402100644N, Date: June 26, 2006
- ✓ **Information Processing Craftsman** – Certification No.: 04402100928U, Date: June 21, 2004

## REFERENCES

References available upon request.